Portable VIS-NIR Spectrometer

DESIGN DOCUMENT

Team SDDEC26-06

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Executive Summary

Provide a brief (300-500 word) summary of your project. This summary should give the reader enough information to understand the purpose, design strategies, and project progress thus far. It should address at least the following questions/items:

- What is the problem and why is it important?
- What are your key design requirements?
- Summarize your design, include approaches and technologies used.
- How much progress have you made thus far?
- How well does your design meet requirements and address user needs? How do you know?
- What are potential next steps to be taken?

The central project that our group is working on is finding a low-cost alternative to professional use spectrometers, especially one in the Near Infrared (NIR). Our motivation for taking on this design challenge is that current spectrometers are expensive, bulky, and not as accessible. We used a high sensitivity linear CCD (TCD1304DG), controlled through a STM32F411CEU6 microcontroller, and we will use a python application to display the data in real-time through USB. Our operating range for this spectrometer is 400nm – 1000nm wavelength. Our current progress includes PCB design verification and fabrication, sourcing optical components, and setting up microcontroller communication. Since our primary focus is on building this for Avishek's person lab first, our metrics for measuring how well we meet the design requirements are quite clear. We have weekly standup meetings where we go over current design implementations, bugs, errors, and general updates on what phase we are in the project. While Avishek has given us room approach this project however we see fit, he makes sure that we are staying within his needs for the project – the operating wavelength, size, and components to use.

Learning Summary

Development Standards & Practices Used

List all standard circuit, hardware, software practices used in this project. List all the Engineering standards that apply to this project that were considered.

Summary of Requirements

List all requirements as bullet points in brief.

Applicable Courses from Iowa State University Curriculum

List all Iowa State University courses whose contents were applicable to your project.

New Skills/Knowledge acquired that was not taught in courses

List all new skills/knowledge that your team acquired which was not part of your Iowa State curriculum in order to complete this project.

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1. Introduction

1.1. Problem Statement

Dr. Avishek Das needs a spectrometer that facilitates his routine experiments in the Biomedical Imaging Lab (BILab) at Iowa State University. In the research environment around the BILab, available spectrometers are expensive, bulky, and effectively fixed to a single location, which can introduce scheduling constraints and delays when an experiment requires measurements in a non-standard setup. These delays reduce iteration speed and limit the ability to validate results.

To address these needs, our project is developing a lower-cost, portable spectrometer that operates across the 400 to 1000-nm wavelength range. The device will operate as a complete package, combining three systems into a compact physical design with the core measurement functions needed for reliable use. First, the instrument will incorporate an optical system that separates incoming light into its component wavelengths, next a sensor and readout system will accurately capture the resulting spectrum, and finally a computer interface will display and record measurement data in a consistent, usable format. This structure will enable us to deliver a self-contained, practical tool that fits Dr. Das's workflows, supports use across the BILab, and remains more affordable than commercially available options.

1.2. Intended Users

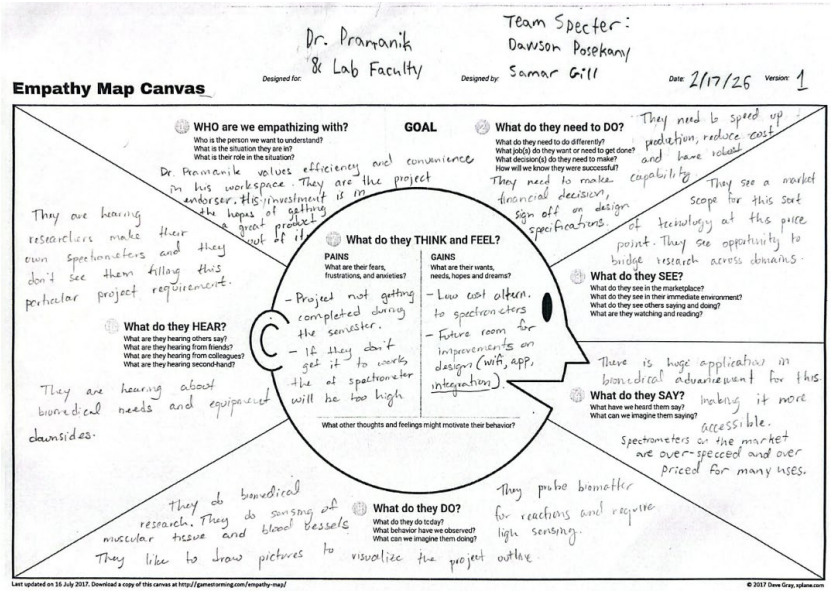
This project was outlined and sponsored by Iowa State researchers at the BILab. Specifically, the project was proposed by Dr. Das under Prof. Pramanik. Through our work with Dr. Das, he has made it clear that he is the only one in the ISU faculty who currently plans on using the device in his research. Because of this, our project will only be considering his needs, although it may be useful to other users as well.

Persona 1: Dr. Das

Dr. Das is a Postdoc Research Associate with the Biomedical Imaging Lab at Iowa State University. He is a highly technical user who leads research in bio-sensing and optics. He is explorative and curious, valuing solutions that do not compromise on portability or efficiency. Encouraged to make his workflow more cost-effective, he attempts to do more with less.

The needs for this device pertain to the research that Dr. Das is doing in biomedicine. The research requires the ability to do quick, non-invasive embedded object imaging in multilayered biological tissues. More specifically, he wants to utilize the characteristics of light at near infrared wavelengths for this imaging. The system needs to be able to be integrated into the rest of his laboratory. This means sending collected information to his computer for analysis and visualization and maybe a phone for modularity. Based on these requirements, Dr. Das needs to quickly and accurately measure the wavelengths of light and easily view the results, while not being restricted by the cost or location of a device.

With the proposed spectrometer, Dr. Das can conduct more frequent and flexible testing. The spectrometer will fit in as a simplified, affordable substitute to solutions currently found on the market. It doesn't come with any extra functionality that won't be used in this kind of research, which maximizes its performance for the specific desired frequencies. The device boosts his lab's throughput and data collection speed.



Note: Dr. Das encouraged publishing this device into scientific literature if the team feels the work warrants it. People outside Dr. Das's lab could then use the device for a variety of purposes. In the way that the team is approaching this project, only the needs of Dr. Avishek and the capabilities of our team are being considered when making design decisions. Due to this, documentation about the needs of external users were placed in the appendix.

2. Requirements, Constraints, And Standards

2.1. Requirements & Constraints

Functional Requirements

- The device shall seperate incoming light into its component wavelengths and capture the resulting spectrum using the TCD1304DG linear CCD sensor.
- The device shall operate across a wavelength range of 400 nm to 1000 nm. *(constraint)*
- The device shall transmit spectral data to a host computer in real time via USB.
- A python application shall display the captured spectrum visually on the host computer.
- The device shall support a configurable integration time to accommodate varying light source intensities.
- The firmware shall drive all required timing signals for the TCD1304DG (master clock, integration clear pulse, shift gate) using the STM32F411CEU6 microcontroller. *(constraint)*

Resource Requirements

- The device shall use the STM32F411CEU6 as the primary microcontroller. *(constraint)*
- The optical subsystem shall use the TCD1304DG linear CCD as the detector. *(constraint)*
- The total bill of material cost shall remain significantly below that of comparable commercial VIS-NIR spectrometers.

Physical Requirements

- The device shall be portable and compact enough to be relocated within a lab environment by a single person without specialized equipment.
- All optical components shall be mountable on a standard optical breadboard using post-mount hardware during the development and testing phase.
- The final device shall include an enclosure that protects internal optics and electronics from ambient light contamination and physical damage.
- The device shall have some protective circuitry to prevent improper use from causing damage.

Performance Requirements

- The device shall achieve sufficient spectral resolution to distinguish features relevant to Dr. Das's biosensing research across the 400-1000 nm range.
- The signal-to-noise ratio of the captured spectrum shall be adequate for reliable identification of spectral peaks under lab lighting conditions.
- The device shall complete a full spectral acquisition and transmission cycle at a rate sufficient for real-time display on the host computer.

User Experience Requirements

- The Python interface shall display a live spectral plot that updates with each new acquisition.
- The device shall be operable by Dr. Das without requiring hardware reconfiguration or adjustments to the code between routine measurements.
- Setup and teardown time for a standard measurement shall be minimal, supporting the fast iteration speed that motivated this project.

Economic Requirements

- The total project cost shall remain within the allocated senior design budget. *(constraint)*
- The final device cost shall be lower than commercially available VIS-NIR spectrometers with comparable wavelength coverage.

Environmental Requirements

- The device shall be designed for indoor laboratory use under standard temperature and humidity conditions.
- Component selection shall prefer parts with reasonable availability and lead times to avoid supply chain delays.
- Materials used shall not be hazardous to people or the environment, assuming proper operation and disposal.

2.2. Engineering Standards

As ethical engineers, our deontological principles dictate that we adhere to IEEE's engineering standards. This team of engineers trust their panel of experts to put out well researched guidelines that will help us ensure our project is safe and reliable.

1. IEC 61010-1: Safety Requirements for Electrical Equipment

- **Standard Type:** Safety
- **Description & Purpose:** This is a foundational safety standard for lab equipment. Its primary goal is to protect the user and environment from hazards like a shock or fire. Crucially, it discusses the single-fault condition which states that even if one component should fail, the device remains safe to touch.
- **Relevance to Vis-NIR Project:** This spectrometer has a light collecting device and a power supply, so the team must adhere to this guideline to ensure the chassis is properly grounded and that any heat generated is efficiently dissipated.
- **Design Modifications:**
    - **Isolation:** Including vias to ground and adequate spacing to reduce danger in the case of a current surge and EMI protection.
    - **Containment:** Making a metal layer between the outer device and the internal optics and electronics with the aim of protecting the user.

2. IEC 61326-1: Electromagnetic Compatibility (EMC) Requirements

- **Standard Type:** Compatibility and Immunity
- **Description & Purpose:** This standard regulates the electromagnetic emissions and immunity of laboratory equipment. It ensures the spectrometer does not interfere with other electronics. On the other hand, it also seeks to ensure that our device can operate accurately in a noisy electromagnetic environment.
- **Relevance to Vis-NIR Project:** Spectrometers rely on highly sensitive photodetectors. The signals that are then produced are relatively weak. Without EMC compliance, external electrical noise could easily distort spectral data, leading to high signal-to-noise ratios. This would ultimately interfere with the high data resolution that we seek to achieve with this device.
- **Design Modifications:**

    - **Shielding:** Including some sort of coating or metal around the analog front-end should protect from outside EM noise.
    - **Filtering:** Decoupling capacitors added to the power traces level out the incoming power and prevent noise from leaking out to other parts of the circuit or off the device as radiation.

3. ASTM E275: Describing and Measuring Performance of UV-Vis Spectrophotometers

- **Standard Type:** Performance and Metrology
- **Description & Purpose:** This standard provides a consistent methodology for verifying the technical performance of a spectrometer. It focuses on characterizing wavelength accuracy, stray light levels, and photometric precision to ensure the data collected is scientifically valid.
- **Relevance to Vis-NIR Project:** This is the benchmark for our project's success. It allows us to compare our "Senior Design" instrument against industry-standard benchmarks. It moves the project from being a "sensor" to a "scientific instrument."
- **Design Modifications:**
    - **Calibration Routine:** Developing a software module that executes a startup self-test, checking the dark current and the reference signal against a known baseline. A device that generates known, overlapping wavelengths will also be used to ensure precision of the sensor.
    - **Opto-mechanical Alignment:** Testing the slit and grating mounts to meet resolution requirements serves us with better data and is in line with this standard. The team will accomplish this by using a variable slit in earlier models for flexibility in the testing phase.

4. NIST-Traceable Wavelength Standards (SRM 2034)

- **Standard Type:** Calibration and Traceability
- **Description & Purpose:** The National Institute of Standards and Technology uploads data from their own devices with standardized absorption peaks. These standards are what all spectrometers are calibrated against.
- **Relevance to Vis-NIR Project:** Calibrating our sensor using an NIST-traceable material is necessary to adhere to code. This ensures that a reading of 900nm on our device is the same as 900nm on any other calibrated device.
- **Design Modifications:**
    - **Reference Holder:** Designing a standardized "sample slot" in the spectrometer housing that can hold a container for the NIST reference material.
    - **Software Correction:** Implementing a "Wavelength Shift" algorithm in the software to adjust the pixel-to-wavelength mapping based on the NIST reference peaks.

3 Project Plan

3.1 Project Management/Tracking Procedures

Earlier on in the project design process, the team implemented a waterfall+agile management style. During this time, the team was ordering most of the parts that would be needed for the remainder of the project, and that led to much more careful steps to ensure that the parts were what was needed for the current design and for future iterations of the design. That said, our other weekly work took a more agile approach where each pair of team members would work on a task or two each week.

Later in the projects' lifecycle, the team took on an agile management strategy. At this point, with our first prototype functioning, it was up to the team to work on improvements, additional features, and integrations with the greater Bio-imaging system Dr. Das is responsible for. The work was more focused on “what can we do next” rather than “what is the next step in this process”.

As for how progress is tracked in this management system, weekly roundups are made during the team’s weekly meetings with Dr. Das. These can be referenced at any point to know what tasks and sub-tasks were created and when. For the software side of our project, a GitLab is hosted which allows us to look at previous points in our development. As for the optics and hardware side of the project, the team stores our files on Microsoft Teams, including manuals, datasheets, and other documentation.

3.2 Task Decomposition

The main tasks involved with designing this project are to set up the optics, generate code for the microcontroller, create a user interface, and set up interaction between different sections of the project. Testing and optimization will also be necessary, as well as a way to calibrate the device.

The optics task can be further broken down into splitting the light based on wavelength, reducing effects of second order diffractions and ambient light, and mapping the spectrum onto the CCD. This will mainly involve selecting parts and determining proper positioning.

For the microcontroller code, it will need to generate the PWM timer signals, include ADC to read the data, use DMA for performance, and implement USB communication with the computer. It will also need to respond to inputs from the computer to determine when to start measurements and change any other parameters.

The user interface will need to graphically display the received data and have options for the user to customize the measurement. This will involve code for USB communication and manipulation of the data to reflect the correct wavelengths and intensities.

In order to put all of these sections together, various smaller tasks will need to be completed. These include selecting and incorporating a battery, creating a housing for the device, and including circuitry for driving the CCD.

3.3 Project Proposed Milestones, Metrics, and Evaluation Criteria

What are some key milestones in your proposed project? It may be helpful to develop these milestones for each task and subtask from 3.2. How do you measure progress on a given task? These metrics, preferably quantifiable, should be developed for each task. The milestones should be stated in terms of these metrics: Machine learning algorithm XYZ will classify with 80% accuracy; the pattern recognition logic on FPGA will recognize a pattern every 1 ms (at 1K patterns/sec throughput). ML accuracy target might go up to 90% from 80%.

In an agile development process, these milestones can be refined with successive iterations/sprints (perhaps a subset of your requirements applicable to those sprint).

Our project is divided into three subsystems (optics, microcontroller and driver board, and computer interface), each of which has a specific design requirement and needs. The milestones below are organized by subsystems and mapped to the quantitative performance targets derived from our requirements.

1. **Subsystem 1: Optical System**: The optical subsystem is responsible for physically separating incoming light into its component wavelengths and directing the resulting spectrum onto the CCD detector.
    1.1. **Milestone 1.1 - Optical Component Procurement and Bench Assembly** All optical components (diffracting grating [transmission grating part number, focusing lens LA1134-AB-ML, slit aperture and mounting hardware) are sourced and assembled on an optical breadboard. *Metric:* All components are physically mounted and aligned within the breadboard footprint, with no missing parts. *Evaluation:* Visual inspection and component checklist.
    1.2. **Milestone 1.2 - Wavelength Coverage Verification** The optical path successfully disperses light and covers the full 400-1000nm operating range. *Metric:* When illuminated by a broadband light source, the CCD pixel array captures a continuous spectrum spanning at least 600 nm (400 to 1000 nm). *Evaluation:* Pixel-to wavelength mapping verified using at least two known spectral emission lines (e.g., from a mercury or neon lamp) at opposite ends of the range.
    1.3. **Milestone 1.3 - Spectral Resolution** The optical system resolves spectral features at a level sufficient for Dr. Das's biosensing work. *Metric:* The full-width at half-maximum (FWHM) of narrow emission line is ≤ 5nm across the 400-1000 nm range. *Evaluation:* A narrow-linewidth source (e.g., laser diode or filtered lamp) is measured and its FWHM is extracted from the acquired spectrum.
2. **Subsystem 2: Microcontroller and Driver Board** The microcontroller and driver board form the electrical backbone of the device. The STM32F411CEU6 must generate precise timing signals to operate the TCD1304DG CCD, perform analog-to-digital conversion of hte pixel data, and transmit results over USB. This subsystem is analogous to the nervous system of the device: everything downstream depends on it firing correctly and on time.
    2.1. **Milestone 2.1 - Firmware Timing Signals Verified** The STM32 firmware correctly drives the three required TCD1304DG timing signals: master clock (fM), integration clear gate (ICG), and shift gate (SH). *Metric:* On an oscilloscope, fM runs at 2MHz (±5%), ICG pulse width ≥1 ms at startup, and SH pulse timing meets the TCD1304DG datasheet specifications. *Evaluation:* Logic analyzer or oscilloscope capture compared against TCD1304DG timing diagram.
    2.2. **Milestone 2.2 - ADC Acquisition of CCD Output** The STM32 ADC acquires the full 3648-pixel CCD output waveform per integration cycle via DMA transfer. *Metric:* A complete frame of 3648 samples is read without dropped samples, confirmed by DMA transfer complete flag and buffer inspection. *Evaluation:* UART or USB debug dump of a single frame; sample count and waveform shape verified.
    2.3. **Milestone 2.3 - USB Data transmission** The device streams spectral frames to the host computer over USB at a rate sufficient for real-time display. *Metric:* Complete 3648-sample frames are transmitted at a rate of ≥10 frames per second at minimum integration time. *Evaluation:* USB packet timing measured on the host; frame rate computed from timestamps.
    2.4. **Milestone 2.5 - PCB Fabrication and Bring-Up** The custom driver PCB is fabricated, assembled, and passes electrical bring-up. *Metric:* All power rails are within ±5% of

nominal voltage, no short circuits, and STM32 boots successfully from flash. *Evaluation:* Multimeter checks on power rails, successful firmware flash and firmware flash and execution confirmed via debug UART.

3. **Subsystem 3: Computer Interface (Python Application)**: The python application serves as the human-facing layer of the spectrometer: it receives raw pixel data from the STM32, applies the wavelength calibration, and presents a live spectral plot. The goal is for Dr. Das to plug in the device and immediately see a spectrum, with no reconfiguration required between routine measurements.
    - 3.1. **Milestone 3.1 - USB Data Reception and Parsing** The Python application successfully receives and parses a full spectral frame from the STM32 over USB. *Metric:* Application receives 3648-sample frames with no data corruption (validated via checksum or frame header/footer markers). *Evaluation:* Captured frames compared to known test patterns transmitted by the firmware.
    - 3.2. **Milestone 3.2 - Live Spectral Display** The application displays a live, updating spectral plot of intensity vs. wavelength *Matric:* Plot updates at ≥5 Hz during continuous acquisition and correctly label the x-axis in nanometers using the calibration mapping. *Evaluation:* Timed observation of display update rate; axis labels verified against calibrated emission line positions.
    - 3.3. **Milestone 3.3 - Dark Current Correction** The application applies a dark frame subtraction to reduce baseline noise. *Metric:* After dark subtractions, the RMS noise floor across a baseline (no-light) acquisition is reduced by at least 50% compared to uncorrected data. *Evaluation:* Statistical comparison of raw vs. dark-corrected frames under identical conditions.
    - 3.4. **Milestone 3.4 - Data Export** The application can save an acquired spectrum to a file for downstream analysis. *Metric:* A spectrum is exportable as a two-column CSV (wavelength in nm, intensity in ADC counts) with a single button press or command. *Evaluation:* Exported file opened in a spreadsheet and columns verified.
4. **System-Level Milestone**
    - 4.1. **Milestone S.1 - Full System Integration and Acceptance** All three subsystems operate together as a single, functional spectrometer. *Metric:* The assembled device, driven by the Python application, acquires and displays a correctly calibrated spectrum form a known reference source, with wavelength error ≤2 nm, FWHM ≤5 nm, and SNR sufficient to clearly resolve at least one identifiable spectral feature. *Evaluation:* Demonstrated live to Dr. Das in a lab session; feedback documented.

3.4 Project Timeline/Schedule

To keep on track and make long term plans, the team made use of a Gantt chart. This allows for the team to manage past progress and evaluate their current status with where they planned to be. The Gantt chart for the first semester can be found below.

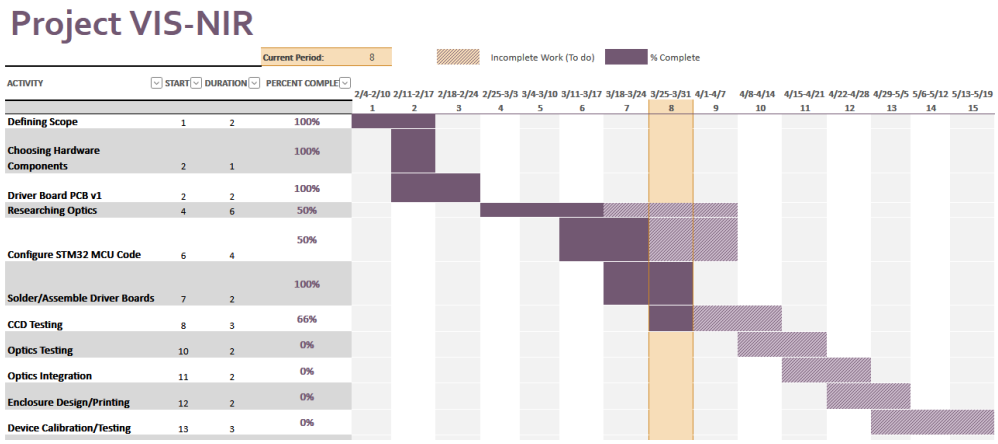


Image 3.4.1: Semester 1 Gantt Chart

The planned schedule had the team ending the first semester with a functioning prototype of the spectrometer, showcasing portability and live spectral analysis. The project fell slightly behind schedule half way through the Spring semester due to the changes that needed to be made to the diffraction grating and filtering.

3.5 Risks and Risk Management/Mitigation

Identified Risk	Probability	Severity	Mitigation strategy
2nd Order Reflections	1.0	Very High	Add linear edge pass filter and validate wavelength mapping experimentally across spectral range. Potential software fixes.
Order Delays	0.5	Medium	Have multiple tasks queued to allow switching tasks while waiting for the other to progress.
Damaging Delicate Optics Components	0.3	High	Wear gloves and handle all optics components according to BILab standards.
Low signal-to-noise ratio at higher wavelengths (800–1000 nm)	0.8	High	Adjustable integration time range, optimize analog front-end gain, and select optics with NIR-optimized coatings.
Limited dynamic range during bright measurements	0.4	Medium	Support adjustable integration timing and optional neutral density filtering.
Optical alignment sensitivity	0.7	High	Use adjustable mounts inside the enclosure.

Enclosure light leakage	0.3	Medium	Add sealing features and verify dark-condition performance during system testing.
Calibration drift over time	0.5	Low	Perform periodic recalibration using known spectral reference sources and store correction tables.
Thermal variation affecting measurements	0.4	Low	Characterize temperature dependence and allow recalibration or compensation in firmware if required.

3.6 Personnel Effort Requirements

Subsystem/Task	Description	Man-hours
Optics		
Overall design	Base-level optics selection including slit, grating, lenses, and mirrors.	10
Second-order light filtering	Evaluating various methods to suppress spectral overlap	20
Component selection	Selection of grating, mirrors, slit, lenses, and optical filter options	12
Testing/Positioning	Alignment testing, wavelength spread verification, and focus optimization	14
Enclosure Design	Designing an enclosure that provides stable mounts between optical components and CCD board	20
Total		**76**
Microcontroller		
MCU IDE Bringup	Familizaring ourselves with the IDE and building the basic code for the CCD driver timers.	12
CCD driver timer implementation	Generating SH, ICG, and master clock timing signals for TCD1304 operation	16
ADC acquisition configuration	Configuring ADC sampling timing synchronization and buffering	10

Testing Basic Code	Verifying CCD waveform timing and signal capture stability	4
Code Improvements	Improve the baseline code to enable inputs from the host pc for calibration modes and selectable integration times.	18
Noise reduction	Implementation of noise reduction processes.	8
Total		**68**
User interface		
Scoping Data Transmission/Features	Validating waveform timing throughput and transfer reliability	8
PC visualization interface	Developing plotting pipeline for spectral display and storage	10
Designing device or selecting filters	Adding selectable integration control, gain selection, and capture triggers	10
Calibration Procedure for pixel map	Mapping pixel index to wavelength using reference emission source	12
Spectral correction workflow	Implementing intensity normalization and dark-frame subtraction	8
Total		**48**
Assembly and Testing		
PCB Design and Assembly	Fabrication of CCD driver PCB and others.	12
Mechanical enclosure assembly	Integrating optics, electronics, and housing into portable structure.	8
System-level integration testing	End-to-end spectrum capture validation across wavelength range and tuning of response.	30
Performance validation	Resolution, SNR, repeatability, and wavelength accuracy testing.	12
Documentation and handoff preparation	Preparing calibration instructions, wiring diagrams, and usage procedures.	8
Total		**70**
Grand Total		**262**

3.7 Other Resource Requirements

Additional resources needed to complete this project (aside from financial assistance) are as follows:

1. Thin film deposition machines available at the ASC will be used to fabricate a linear edge pass filter that will be used to filter 2nd order reflections.

4 Design

4.1 Design Context

4.1.1 Broader Context

Describe the broader context in which your design problem is situated. What communities are you designing for? What communities are affected by your design? What societal needs does your project address?

List relevant considerations related to your project in each of the following areas:

Area	**Description**	**Examples**
Public health, safety, and welfare	How does your project affect the general well-being of various stakeholder groups? These groups may be direct users or may be indirectly affected (e.g., solution is implemented in their communities)	Increasing/reducing exposure to pollutants and other harmful substances, increasing/reducing safety risks, increasing/reducing job opportunities
Global, cultural, and social	How well does your project reflect the values, practices, and aims of the cultural groups it affects? Groups may include but are not limited to specific communities, nations, professions, workplaces, and ethnic cultures.	Development or operation of the solution would violate a profession's code of ethics, implementation of the solution would require an undesired change in community practices
Environmental	What environmental impact might your project have? This can include indirect effects, such as deforestation or unsustainable practices related to materials manufacture or procurement.	Increasing/decreasing energy usage from nonrenewable sources, increasing/decreasing usage/production of non-recyclable materials
Economic	What economic impact might your project have? This can include the financial viability of your product within your team or company, cost to consumers, or broader economic effects on communities, markets, nations, and other groups.	Product needs to remain affordable for target users, product creates or diminishes opportunities for economic advancement, high development cost creates risk for organization

4.1.2 Prior Work/Solutions

Include relevant background/literature review for the project (cite at least 3 references for literature review in IEEE Format. See link: https://ieee-dataport.org/sites/default/files/analysis/27/IEEE%20Citation%20Guidelines.pdf)

– If similar products exist in the market, describe what has already been done

– If you are following previous work (e.g., a previous senior design project), cite that and discuss the **advantages/shortcomings**

– Note that while you are not expected to "compete" with other existing products / research groups, you should be able to differentiate your project from what is available. Thus, provide a list of pros and cons of your target solution compared to all other related products/systems.

Detail any similar products or research done on this topic previously. Please cite your sources and include them in your references. All figures must be captioned and referenced in your text.

4.1.3 Technical Complexity

Provide evidence that your project is of sufficient technical complexity. Use the following metric or argue for one of your own. Justify your statements (e.g., list the components/subsystems and describe the applicable scientific, mathematical, or engineering principles)

1. The design consists of multiple components/subsystems that each utilize distinct scientific, mathematical, or engineering principles –AND–
2. The problem scope contains multiple challenging requirements that match or exceed current solutions or industry standards.

4.2 Design Exploration

4.2.1 Design Decisions

List key design decisions (at least three) that you have made or will make. These can include, but are not limited to, materials, subsystems, physical components, sensors/chips/devices, physical layout, features, etc. Describe how you made/will make these decisions and how they have affected or are likely to affect project success.

4.2.2 Ideation

For at least one design decision, describe how you ideated or identified potential options (e.g., lotus blossom technique). Describe at least five options that you considered.

4.2.3 Decision-Making and Trade-Off

Demonstrate the process you used to identify the pros and cons or trade-offs between each of your ideated options. You may wish you include a weighted decision matrix or other relevant tool. Describe the option you chose and why you chose it.

4.3 Proposed Design

4.3.1 Overview

Provide a high-level description of your current design. This description should be understandable to non-engineers (i.e., the general public). Describe key components or sub-systems and how they contribute to the overall design. You may wish to include a basic block diagram, infographic, or other visual to help communicate the overall design.

4.3.2 Detailed Design and Visual(s)

Provide a detailed, technical description of your design, aided by visualizations. This description should be understandable to peer engineers. In other words, it should be clearly written and sufficiently detail such that another senior design team can look through it and implement it.

The description should include a high-level overview written for peer engineers. This should list all sub-systems or components, their role in the whole system, and how they will be integrated or interconnected. A visual should accompany this description. Typically, a detailed block diagram will suffice, but other visual forms can be acceptable.

The description should also include more specific descriptions of sub-systems and components (e.g., their internal operations). Once again, a good rule of thumb is: could another engineer with similar expertise build the component/sub-system based on your description? Use visualizations to support your descriptions. Different visual types may be relevant to different types of projects, components, or subsystems. You may include, but are not limited to: block diagrams, circuit diagrams, sketches/pictures of physical components and their operation, wireframes, etc.

4.3.3 Functionality

Describe how your design is intended to operate in its user and/or real-world context. What would a user do? How would the device/system/etc. respond? This description can be supplemented by a visual, such as a timeline, storyboard, or sketch.

4.3.4 Areas of Concern and Development

How well does/will the current design satisfy requirements and meet user needs?

Based on your current design, what are your primary concerns for delivering a product/system that addresses requirements and meets user and client needs?

What are your immediate plans for developing the solution to address those concerns? What questions do you have for clients, TAs, and faculty advisers?

4.4 Technology Considerations

Describe the distinct technologies you are using in your design. Highlight the strengths, weakness, and trade-offs made in technology available. Discuss possible solutions and design alternatives.

4.5 Design Analysis

Discuss what you have done so far, i.e., what have you built, implemented, or tested? Did your proposed design from 4.3 work? Why or why not? Based on what has worked or not worked (e.g., what you have or haven't been able to build, what functioned as expected or not), what plans do you have for future design and implementation work? For example, are there implications for the overall feasibility of your design or have you just experienced build issues?

5 Testing

Testing is an **extremely** important component of most projects, whether it involves a circuit, a process, power system, or software.

The testing plan should connect the requirements and the design to the adopted test strategy and instruments. In this overarching introduction, give an overview of the testing strategy and your team's overall testing philosophy. Emphasize any unique challenges to testing for your system/design.

In the sections below, describe specific methods for testing. You may include additional types of testing, if applicable to your design. If a particular type of testing is not applicable to your project, you must justify why you are not including it.

When writing your testing planning consider a few guidelines:

- Is our testing plan unique to our project? (It should be)
- Are you testing related to all requirements? For requirements you're not testing (e.g., cost related requirements) can you justify their exclusion?
- Is your testing plan comprehensive?
- When should you be testing? (In most cases, it's early and often, not at the end of the project)

5.1 Unit Testing

What units are being tested? How? Tools?

5.2 Interface Testing

What are the interfaces in your design? Discuss how the composition of two or more units (interfaces) are being tested. Tools?

5.3 Integration Testing

What are the critical integration paths in your design? Justification for criticality may come from your requirements. How will they be tested? Tools?

5.4 System Testing

Describe system level testing strategy. What set of unit tests, interface tests, and integration tests suffice for system level testing? This should be closely tied to the requirements. Tools?

5.5 Regression Testing

How are you ensuring that any new additions do not break the old functionality? What implemented critical features do you need to ensure do not break? Is it driven by requirements? Tools?

5.6 Acceptance Testing

How will you demonstrate that the design requirements, both functional and non-functional are being met? How would you involve your client in the acceptance testing?

5.7 Security Testing (if applicable)

5.8 User Testing

Describe how the team will include real users to interact with the solution/product. Your user testing plan should evaluate your solution/product's usability and overall user experience by gathering user feedback through observations, surveys, etc. User feedback allows the team to identify areas for improvement and ensure the product meets user needs.

5.9 Results

What are the results of your testing thus far? Include any numerical, graphical, or qualitative testing results here? How do they demonstrate compliance with the requirements or addressing user needs? Use a summary narrative to discuss what you've learned and what next steps need to be taken.

6 Implementation

Describe any (preliminary) implementations of your design thus far. Support any general, descriptive text with relevant images. If your project has inseparable activities between design and implementation, you can list them either in the Design section or this section.

7 Ethics and Professional Responsibility

Use this section to describe your considerations of engineering ethics and professional responsibility. Most importantly how are you defining engineering ethics and professional responsibility in the context of your project and what steps are you taken to ensure ethical and responsible conduct. Each section references one type of ethical/professional responsibility considerations. You may also use this introductory section to note any overarching ethical philosophy among your team.

7.1 Areas of Professional Responsibility/Codes of Ethics

This discussion is with respect to the paper by J. McCormack and colleagues titled "Contextualizing Professionalism in Capstone Projects Using the IDEALS Professional Responsibility Assessment", *International Journal of Engineering Education* Vol. 28, No. 2, pp. 416–424, 2012

Pick one of IEEE, ACM, or SE code of ethics (all linked in class slides). Create a table, like Table 1 in the McCormack et al. (2012, pg. 418)) paper, with the following columns representing: Area of Responsibility (from the paper), Definition (in your own words), Relevant Item from Code of Ethics (from the Code of Ethics you selected, and description of how your team has interacted with that area of professional responsibility or adhered to that code during your project thus far.

In text below the table, describe one area in which your team is performing well. Describe what your team is doing and how that signifies strong performance. Also describe one area in which your team needs to improve. Describe what your team is currently doing and what it should do in the future to improve.

7.2 Four Principles

Create a table with rows for each broader context area (see Section 4.1.1) and columns for each of the four principles (beneficence, nonmalificence, respect for autonomy, and justice; see Beauchamp, 2007). Within the table, identify at least one way each of the four principles applies to each of the broader context areas. Some principle-broader context connections might be more prominent than others, but you should be able to identify something for each table cell. Note: Your design may end up negative or neutral in some cell. For example, your product might perform poorly in environment-nonmaleficence because it utilizes natural resources without a positive/mitigating effect.

Below the table, note one broader context-principle pair that is important to your project. Briefly describe the benefit in that area you are working towards and how you will ensure it. Also note one broader context-principle pair in which your project/end design is or will be lacking. Describe either (a) how this negative is overcome by other positives in other areas of the project/design or (b) what your team must do to improve in this area.

7.3 Virtues

List and define at least three virtues that are important to your team. Describe what you will do or have done as a team to support these virtues among all team members.

Each team member should also answer the following:

- Identify one virtue you have demonstrated in your senior design work thus far? (Individual)
    - Why is it important to you?
    - How have you demonstrated it?
- Identify one virtue that is important to you that you have not demonstrated in your senior design work thus far? (Individual)
    - Why is it important to you?
    - What might you do to demonstrate that virtue?

8 Closing Material

8.1 Conclusion

Summarize the work you have done so far. Briefly re-iterate your goals. Then, re-iterate the best plan of action (or solution) to achieving your goals. What constrained you from achieving these goals (if something did)? What could be done differently in a future design/implementation iteration to achieve these goals?

8.2 References

List technical references and related work / market survey references. Do professional citation style (ex. IEEE). See link: https://ieee-dataport.org/sites/default/files/analysis/27/IEEE%20Citation%20Guidelines.pdf

8.3 Appendices

Personas we were considering

Any additional information that would be helpful to the evaluation of your design document.

If you have any large graphs, tables, or similar data that does not directly pertain to the problem but helps support it, include it here. This would also be a good area to include hardware/software manuals used. May include CAD files, circuit schematics, layout etc,. PCB testing issues etc., Software bugs etc.

Persona 1: Biomedical/Electrical Engineering Researcher

- **Demographic:** Adult, Researcher, PhD
- **Hobbies:** Learning
- **Motivation:** Money, Scientific advancement
- **Personality:** Passionate, Curious

- **Values:** Accuracy, understanding of the world

This user is a professional in their respective field of study. They are driven by a desire for scientific advancement. They are well versed in each of the fields of work used to make this device, including optics, embedded systems, and analog circuitry. They are highly technical and value precision and efficiency. The researcher needs access to an accurate yet affordable spectrometer that can be used in various applications. This tool offers a way to streamline testing, ultimately speeding up the researchers' contribution to the scientific community.

Persona 2: Medical Staff

- **Demographic:** Specialized Staff in bio-sensing
- **Hobbies/Interests:** Passionate workers care and keep up to date with new bio-sensing technology.
- **Work Motivations:** Making testing and data collection intuitive and usable for more people.
- **Values:** They value efficiency and reliability to draw quick, accurate conclusions.
- **Personality/Emotions:** They want the best for their clients and fellow faculty

This user is a specialized medical staff member with a focus on bio-sensing technology. Staying current with the latest developments in the field is both a professional priority and a personal interest. This user is deeply motivated by making testing and data collection more intuitive and accessible, not just for specialists but for a wider range of clinical staff.

Efficiency and reliability are central to how this user works. In a medical setting, quick and accurate conclusions are not a luxury; they directly impact the quality of care. This user wants tools that are dependable and easy to interpret and genuinely care about the wellbeing of both clients and colleagues.

A portable, accessible spectrometer aligns well with this user's goals. The ability to perform non-invasive optical measurements quickly and reliably could streamline workflows that currently require more complex or time-consuming methods. For this user, the value lies in a tool that brings meaningful sensing capability into everyday clinical use without demanding deep technical expertise to operate.

Persona 3: Educational Institutions

- **Demographic:** High school or University Lab Instructor.
- **Hobbies/Interests:** Science Communication, niche design projects.
- **Work Motivations:** Bridging theory and hands on learning for students.
- **Values:** They need cheap and portable devices so that they can supply a classroom. The devices must be durable and “student-proof”. Encapsulation is crucial.
- **Personality/Emotions:** They are encouraging and patient, wanting to share the natural world with their students.

9 Team

Complete each section as completely and concisely as possible. We strongly recommend using tables or bulleted lists when applicable.

9.1 Team Members

9.2 Required Skill Sets for Your Project

(if feasible – tie them to the requirements)

9.3 Skill Sets covered by the Team

(for each skill, state which team member(s) cover it)

9.4 Project Management Style Adopted by the team

Typically, Waterfall or Agile for project management.

9.5 Initial Project Management Roles

(Enumerate which team member plays what role)

9.6 Team Contract

Team Members:

1) _____ 2) _____

3) _____ 4) _____

5) _____ 6) _____

7) _____ 8) _____

Team Procedures

Day, time, and location (face-to-face or virtual) for regular team meetings:

2. Preferred method of communication updates, reminders, issues, and scheduling (e.g., e-mail, phone, app, face-to-face):

3. Decision-making policy (e.g., consensus, majority vote):

4. Procedures for record keeping (i.e., who will keep meeting minutes, how will minutes be shared/archived):

Participation Expectations

1. Expected individual attendance, punctuality, and participation at all team meetings:

2. Expected level of responsibility for fulfilling team assignments, timelines, and deadlines:

3. Expected level of communication with other team members:

4. Expected level of commitment to team decisions and tasks:

Leadership

1. Leadership roles for each team member (e.g., team organization, client interaction, individual component design, testing, etc.):

2. Strategies for supporting and guiding the work of all team members:

3. Strategies for recognizing the contributions of all team members:

Collaboration and Inclusion

1. Describe the skills, expertise, and unique perspectives each team member brings to the team.

2. Strategies for encouraging and support contributions and ideas from all team members:

3. Procedures for identifying and resolving collaboration or inclusion issues (e.g., how will a team member inform the team that the team environment is obstructing their opportunity or ability to contribute?)

Goal-Setting, Planning, and Execution

1. Team goals for this semester:

2. Strategies for planning and assigning individual and team work:

3. Strategies for keeping on task:

Consequences for Not Adhering to Team Contract

1. How will you handle infractions of any of the obligations of this team contract?

2. What will your team do if the infractions continue?

****

a) I participated in formulating the standards, roles, and procedures as stated in this contract.

b) I understand that I am obligated to abide by these terms and conditions.

c) I understand that if I do not abide by these terms and conditions, I will suffer the consequences as stated in this contract.

1) _____ DATE _____

2) _____ DATE _____

3) _____ DATE _____

4) _____ DATE _____

5) _____ DATE _____

6) _____ DATE _____

7) _____ DATE _____

8) _____ DATE _____